

§4.1 End-effector velocity

We know how to calculate the whole path $g_{st}(\theta(t)) \in SE(3)$ of the End-effector as its joints move along a path $\theta(t) \in Q$.

The instantaneous spatial velocity of the End-effector can then be calculated as the twist

$$\hat{V}_{st}^s = \dot{g}_{st}(\theta)g_{st}^{-1}(\theta)$$

(using eq. (2.53) p. 54).

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$$\hat{V}_{st}^s = \dot{g}_{st}(\theta) g_{st}^{-1}(\theta)$$

Using the chain rule for differentiation

$$\dot{g}_{st}(\theta) = \sum_{i=1}^n \frac{\partial g_{st}(\theta)}{\partial \theta_i} \frac{\partial \theta_i}{\partial t} = \sum_{i=1}^n \frac{\partial g_{st}(\theta)}{\partial \theta_i} \dot{\theta}_i$$

$$\hat{V}_{st}^s = \left(\sum_{i=1}^n \frac{\partial g_{st}(\theta)}{\partial \theta_i} \dot{\theta}_i \right) g_{st}^{-1}(\theta) = \sum_{i=1}^n \left(\frac{\partial g_{st}(\theta)}{\partial \theta_i} g_{st}^{-1}(\theta) \right) \dot{\theta}_i \quad (3.51)$$

Considering eq. (2.53) p. 54,

$$\left(\frac{dg_{st}}{dt} g_{st}^{-1} \right) \text{ is a twist. Is } \left(\frac{\partial g_{st}(\theta)}{\partial \theta_i} g_{st}^{-1}(\theta) \right) \text{ a twist?}$$

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$$\widehat{V}_{st}^s = \sum_{i=1}^n \left(\frac{\partial g_{st}(\theta)}{\partial \theta_i} g_{st}^{-1}(\theta) \right) \dot{\theta}_i \quad (3.51)$$

Because $\left(\frac{\partial g_{st}(\theta)}{\partial \theta_i} g_{st}^{-1}(\theta) \right)$ is indeed a twist, we can write

$$V_{st}^s = J_{st}^s(\theta) \dot{\theta} ,$$

$$\text{where } J_{st}^s(\theta) = \left[\left(\frac{\partial g_{st}}{\partial \theta_1} g_{st}^{-1} \right)^\vee \cdots \left(\frac{\partial g_{st}}{\partial \theta_n} g_{st}^{-1} \right)^\vee \right] \quad (3.52)$$

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$$V_{st}^s = J_{st}^s(\theta)\dot{\theta}$$

$V_{st}^s \in \mathbb{R}^{6 \times n}$ — spatial manipulator Jacobian

At each configuration θ it maps the joint velocity vector into the corresponding velocity of the End-effector.

$$\begin{aligned} J_{st}^s(\theta) &= \begin{bmatrix} \xi_1 & \xi'_2 \cdots \xi'_n \end{bmatrix} \\ \xi'_i &= \text{Ad}_{g_{s,i-1}} \xi_i \\ g_{s,i-1} &= e^{\hat{\xi}_1 \theta_1} \cdots e^{\hat{\xi}_{i-1} \theta_{i-1}} \end{aligned} \tag{3.54}$$

The i -th column of the spatial Jacobian is the i -th joint twist, transformed to the current manipulator configuration.

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Spatial manipulator Jacobian J_{st}^s : $V_{st}^s = J_{st}^s(\theta)\dot{\theta}$

Body manipulator Jacobian J_{st}^b : $V_{st}^b = J_{st}^b(\theta)\dot{\theta}$

$$\begin{aligned} J_{st}^b(\theta) &= \begin{bmatrix} \xi_1^\dagger & \xi_2^\dagger & \cdots & \xi_n^\dagger \end{bmatrix} \\ \xi_i^\dagger &= \text{Ad}_{g_{i,t}}^{-1} \xi_i \\ g_{i,t} &= e^{\hat{\xi}_i \theta_i} \cdots e^{\hat{\xi}_n \theta_n} g_{st}(0) \end{aligned} \tag{3.55}$$

The i -th column of the body Jacobian is the i -th joint twist, written with respect to the tool frame at the current manipulator configuration.

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$$J_{st}^s(\theta) = \text{Ad}_{g_{st}(\theta)} J_{st}^b(\theta) \quad (3.56)$$

For q^b the coordinates in the tool frame of a point on the End-effector, using eq. (2.56) p. 55:

$$v_q^b = \widehat{V}_{st}^b q^b = \left(J_{st}^b(\theta) \dot{\theta} \right)^\wedge q^b$$

For q^s the coordinates in the spatial frame of a point on the End-effector, using eq. (2.54) p. 54:

$$v_q^s = \widehat{V}_{st}^s q^s = \left(J_{st}^s(\theta) \dot{\theta} \right)^\wedge q^s$$

§4.2 End-effector forces

Seeking a relationship between the wrench applied to the end-effector and the joint torques

$$\int_{t_1}^{t_2} \dot{\theta}^T \tau \, dt = W = \int_{t_1}^{t_2} V_{st}^{bT} F_t \, dt$$

$$\int_{t_1}^{t_2} \dot{\theta}^T \tau \, dt = W = \int_{t_1}^{t_2} \dot{\theta}^T J_{st}^{bT} F_t \, dt$$

$$\dot{\theta}^T \tau = \dot{\theta}^T J_{st}^{bT} F_t$$

$$\tau = J_{st}^{bT} F_t \quad (3.59)$$

$$\tau = J_{st}^{sT} F_s \quad (3.60)$$

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$$\tau = J_{st}^{bT} F_t \quad (3.59)$$

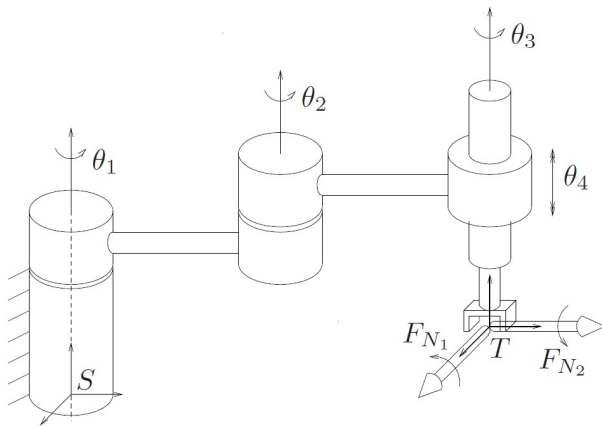
$$\tau = J_{st}^{sT} F_s \quad (3.60)$$

These answer the question: for applied wrench on the end effector, what are the torques τ to resist the applied wrench.

It does not generally answer the question “what torques τ must be applied to cause the end effector to apply a certain required wrench”.

(carefully study MLS p. 122)

§4.2 End-effector forces: Ex 3.10 p. 122



The reference configuration of the robotic arm shown in a fig. 3.15 p. 118 (Ex. 3.8)

§4.2 End-effector forces: Ex. 3.10 p. 122

From Ex. 3.8 p. 118:

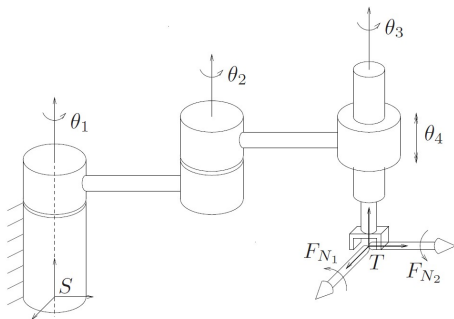
$$J_{st}^s = \begin{bmatrix} 0 & l_1 \cos \theta_1 & l_1 \cos \theta_1 + l_2 \cos(\theta_1 + \theta_2) & 0 \\ 0 & l_1 \sin \theta_1 & l_1 \sin \theta_1 + l_2 \sin(\theta_1 + \theta_2) & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

$$(J_{st}^s)^T = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ l_1 c_1 & l_1 s_1 & 0 & 0 & 0 & 1 \\ l_1 c_1 + l_2 c_{12} & l_1 s_1 + l_2 s_{12} & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

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$$(J_{st}^s)^T = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ l_1 c_1 & l_1 s_1 & 0 & 0 & 0 & 1 \\ l_1 c_1 + l_2 c_{12} & l_1 s_1 + l_2 s_{12} & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

$$F_{N_1} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

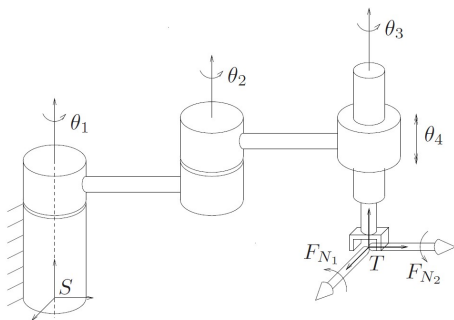


Tool wrench $F_{N_1} \in \text{null space of } (J_{st}^s)^T$, generates no joint torques τ ; no motion

§4.2 End-effector forces: Ex. 3.10 p. 122

$$(J_{st}^s)^T = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ l_1 c_1 & l_1 s_1 & 0 & 0 & 0 & 1 \\ l_1 c_1 + l_2 c_{12} & l_1 s_1 + l_2 s_{12} & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

$$F_{N_2} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$



Similarly, tool wrench $F_{N_2} \in \text{null space of } (J_{st}^s)^T$, generates no joint torques τ ; no motion

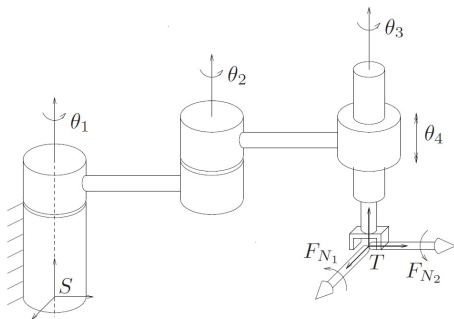
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$$(J_{st}^s)^T = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ l_1 c_1 & l_1 s_1 & 0 & 0 & 0 & 1 \\ l_1 c_1 + l_2 c_{12} & l_1 s_1 + l_2 s_{12} & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

But is this a valid conclusion?

$$\tau = (J_{st}^b)^T F_t$$

$$\tau = (J_{st}^s)^T F_s$$



Similarly, tool wrench $F_{N_2} \in \text{null space of } (J_{st}^s)^T$, generates no joint torques τ ; no motion

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It can be shown that

- ▶ both tool wrenches F_{N_1} & $F_{N_2} \in$ null space of $(J_{st}^b)^T$, generate no joint torques τ ; no motion
- ▶ both tool wrenches $F_{N_1}^s$ & $F_{N_2}^s \in$ null space of $(J_{st}^s)^T$, generate no joint torques τ ; no motion